

BDC60 Brushed DC Gearmotor

SKU / Model Number: BDC60-24-50

Manufacturer: MicroDrive Systems

Category: Motors | ETIM Code: EC001835

PRODUCT OVERVIEW

The BDC60 is a compact brushed DC motor with integrated planetary gearbox for conveyor systems, AGVs, and automated guided vehicles. Features encoder feedback and reversible operation.

TECHNICAL SPECIFICATIONS

Rated Power	60 W
Gear Ratio	50:1
Output Torque	8 Nm (continuous)
Output Speed	60 RPM
No-load Speed	72 RPM
Stall Torque	24 Nm
Efficiency	78%

ELECTRICAL RATINGS

Rated Voltage	24 V DC
No-load Current	0.5 A
Rated Current	3.2 A
Stall Current	9.5 A
Encoder	512 PPR, quadrature

DIMENSIONS & MECHANICAL

Motor Diameter	60 mm
Gearbox Diameter	52 mm
Total Length	148 mm
Shaft Diameter	8 mm
Weight	850 g

OPERATING CONDITIONS

Temperature Range	-10 to 50 deg C
Humidity	10 to 90% RH
Duty Cycle	S1 Continuous
Brush Life	> 3000 hours
IP Rating	IP54

MATERIALS OF CONSTRUCTION

Motor Housing	Zinc alloy die-cast
Gears	Hardened steel
Shaft	Stainless Steel 303
Brushes	Precious metal alloy
Bearings	Sintered bronze

PERFORMANCE DATA / SELECTION TABLE

Gear Ratio	Speed (RPM)	Torque (Nm)	Current (A)
10:1	300	1.6	3.2
20:1	150	3.2	3.2
50:1	60	8.0	3.2
100:1	30	15.0	3.2

CERTIFICATIONS & APPROVALS

CE | UL | RoHS | REACH

Note: Specific certification documents and test reports are available upon request. All products are manufactured under ISO 9001:2015 quality management system.

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